Flowcharts

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Complete framework

1. **Drone video, signal timing, geometry**
Mixed traffic at signalized approaches, georeferenced coordinates

2. **Validated trajectory database**
Motorcycles, cars, buses, trucks

3. **Objective 1**
Interaction and lane-filtering criteria
2D offsets · safety envelope · maneuver model

4. **Objective 2**
Safety evaluation and index
TTC · PET · DRAC · thresholds · safety index

5. **Objective 3**
Behavior model in SUMO
Hybrid model · baseline sub-lane model comparison · validation

6. **Results**
Validated SUMO motorcycle model performance with the baseline sub-lane model, plus thresholds identifying unsafe maneuvers

Objective 1: Interaction, safety envelope and filtering criteria

1. **Select sites and collect data**
Signalized approaches; drone video, signal timing, geometry, reference distances

2. **Extract and validate trajectories**
Detect, track, classify using DFS; georeferenced with ground-control points; smoothing

3. **Compute kinematics and 2D interactions**
Speed, acceleration, heading, Cx, Cy, C2D, edge-to-edge clearance, neighbors

4. **Calibrate elliptical safety envelope**
Speed-dependent ellipse; Envelope Index

5. **Identify maneuvers**
Filtering, lateral shift, overtaking, swerving, following, queue jumping, stop-and-go; verify on video

6. **Extract maneuver variables**
Lateral gap, speed differential, neighbor type, density, queue position, signal phase

7. **Model filtering criteria**
Filtering vs non-filtering comparison; multinomial logit for maneuver probabilities

8. **Outputs**
trajectory database, 2D dataset, calibrated envelope, EI, maneuver labels, probability model.

Objective 2: Safety evaluation, thresholds and safety index

1. **Input from Objective 1**
trajectories, maneuver labels, envelope

2. **Select interaction events**
Motorcycle with cars, buses, trucks, other motorcycles

3. **Compute surrogate safety measures**
2D TTC, PET, DRAC, minimum lateral clearance, relative speed

4. **Add envelope indicators**
EI, minimum EI, penetration depth, time inside envelope

5. **Set percentile thresholds**
Low percentiles (5th) for TTC, PET, clearance; high (95th) for DRAC; tested on held-out data

6. **Build composite safety index**
Normalize; PCA weights; bootstrap resampling

7. **Validate manually on video**
Independent reviews on a subset; report agreement

8. **Classify severity and compare maneuvers**
Percentile cutoffs (relative severity); by maneuver, signal phase, speed, vehicle type

9. **Outputs**
SSM thresholds, safety index, validated severity classes, maneuver-specific comparisons.

Objective 3: Motorcycle behavior model in SUMO

1. **Extract trajectories from DFS**
Position, speed, acceleration, heading, lane position, surrounding-vehicle information

2. **Identify motorcycle maneuvers**
Filtering, following, lateral overtaking, swerving, merging, positioning near the stop line

3. **Develop three behavior components**
Estimate 2D interactions, safety envelopes and probabilistic maneuver choices

4. **Integrate longitudinal and lateral behavior**
Combine car-following (e.g. IDM) with motorcycle-specific lateral movement

5. **Implement the model in SUMO**
Baseline sub-lane model as comparator; custom controller (Python/TraCI)

6. **Validate against independent DFS trajectories**
Compare observed and simulated positions, speeds, gaps, maneuvers and interactions

7. **Evaluate performance**
Does the proposed model reproduce real motorcycle behavior more accurately than the baseline SUMO model?

8. **Outputs**
hybrid motorcycle behavior model in SUMO; validation and comparison against the baseline sub-lane model.